Question	Answer	Marks
1	**Defining the problem**	
	vary v and measure I **or** v is the independent variable and I is the dependent variable	**1**
	keep A and R constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • fan positioned in line with the turbine so that blades of both fan and turbine overlap • fan on bench • fan labelled and one other label from bench, turbine, blade(s) (of turbine), terminals, L	**1**
	workable circuit diagram showing resistor connected to an ammeter in series with the terminals of the turbine using correct circuit symbols	**1**
	method to vary v, e.g. change speed of fan / change distance between fan and blades / vary current in or p.d. across fan	**1**
	method to determine temperature T, e.g. use a thermometer	**1**
	Method of Analysis	
	plot a graph of I^2 against v^3 or equivalent (e.g. lg I against lg v or 2 ln I against ln v)	**1**
	relationship valid if a straight line is produced passing through the origin (For lg I against lg v: relationship valid if a straight line with gradient = 1.5 is produced)	**1**
	$Q = \frac{2TR \times \text{gradient}}{AP}$ (For lg I against lg v: $Q = \frac{2TR \times 10^{2 \times y\text{-intercept}}}{AP}$)	**1**

Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 precaution with reason linked to (moving) fan blades / turbine blades, e.g. keep away from the fan to avoid (moving) blades **or** use a screen around the fan / turbine to avoid (moving) blades **or** precaution with reason linked to prevent air / dust particles in eye, e.g. use goggles to avoid air stream (into eye)	
	D2 clamp turbine / fan to bench	
	D3 keep P and T constant	
	D4 $T = t + 273$	
	D5 method to determine A: use a rule(r) / calipers to measure L **and** $A = \pi L^2$	
	D6 repeat measurements of L in different positions / different blades **and** average	
	D7 method to measure v, e.g. use an anemometer or air speed meter **or** method to measure P, e.g. use a manometer or barometer or pressure gauge	
	D8 wait for steady / constant air flow / movement of blades / current	
	D9 method to determine R, e.g.: separate circuit showing ohmmeter connected to R only **or** terminals of turbine connected correctly to resistor and ammeter and voltmeter across resistor **and** $R = V / I$ **or** separate workable circuit with power supply resistor, ammeter and voltmeter across R **and** $R = V / I$	
	D10 method to check temperature / pressure is constant, e.g. measure temperature / pressure several times / before and after	

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